Session 27: Employee Benefits: Pension Accounting (IAS19)

By the end of this session you must know how IAS19 classifies the four categories of employee benefits, the difference between defined contribution and defined benefit plans (and who bears actuarial/investment risk under each), how to compute the defined benefit obligation using the projected unit credit method, and how to split the year's total remeasurement between profit or loss and OCI.

Plan

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1. IAS19's Four Categories of Employee Benefits

Employee benefits (IAS19): all forms of consideration given by an entity in exchange for service rendered by employees, or for the termination of employment.

Must know (the four categories, how measured, when recognised):

• **Short-term benefits** (wages, salaries, bonuses, paid leave, staff benefits): expected to be settled within 12 months of the reporting period end. Recognised as employees provide service, at the undiscounted amount expected to be paid

• **Post-employment benefits** (pensions, retirement medical plans, gratuities): recognised as employees provide service, but payable after employment ends. Measured either as defined contribution (expense contributions when due) or defined benefit (actuarial valuation + discounting), the focus of this session

• **Other long-term benefits** (long-service leave, long-term bonuses): recognised as employees provide service but payable more than 12 months after the service period ends. Measured similarly to defined benefit plans, **except** all remeasurement changes go to profit or loss (none to OCI)

• **Termination benefits** (redundancy/severance pay): recognised when the entity can no longer withdraw the offer of those benefits (or when it recognises related restructuring costs), recognise a liability for the expected payment

2. Defined Contribution Plans

Post-employment benefit plan: a formal or informal arrangement under which an entity provides post-employment benefits to employees, classified as either defined contribution or defined benefit based on the economic substance of its terms.

Defined contribution plan: the entity pays fixed contributions into a separate fund and has no legal or constructive obligation to pay further contributions if the fund's assets prove insufficient to pay the promised benefits.

- Because the employee's eventual benefit depends entirely on the contributions paid plus the investment returns they generate, both actuarial risk (benefits turning out lower than expected) and investment risk (assets proving insufficient) fall, in substance, on the **employee**, not the entity.
- Accounting is simple: contributions payable during the period are expensed. Any unpaid contribution liability at period-end is recognised (and discounted if due more than 12 months after the period in which the employee provided the related service).

3. Defined Benefit Plans: Who Bears the Risk

Defined benefit plan: any post-employment benefit plan that is not a defined contribution plan. The entity's obligation is to provide the agreed benefits to current and former employees, and actuarial and investment risk fall, in substance, on the **entity**: if actuarial or investment experience is worse than expected, the entity's obligation increases.

- This is the mirror image of Section 2: under defined benefit, the employer (not the employee) absorbs the risk that the pension promise costs more than originally estimated.

4. The Defined Benefit Obligation: Measurement Principles

Must know (the defined benefit accounting model, four building blocks):
• The balance-sheet **liability** is estimated using the **projected unit credit method**, an actuarial technique estimating the final cost of benefits accrued for current and prior years' service, based on demographic assumptions (turnover, mortality) and financial assumptions (future salary increases, medical cost inflation), discounted to present value
• That liability is **reduced by plan assets** (measured at fair value). Funds set aside (often in securities) to meet future benefit payments. If the obligation exceeds plan assets, there is a **deficit** (net liability). If plan assets exceed the obligation, a **surplus** (net asset)
• The **P&L cost** comprises: service cost (cost of the current year's work) plus **net interest** (interest cost on the obligation from the passage of time, net of interest income on plan assets, both using the **same discount rate**)
• **Remeasurements** (actuarial gains/losses from changes in discount rate, turnover/mortality estimates, salary-increase assumptions, or from the gap between actual and expected plan-asset returns) go immediately to **OCI** and are **never recycled** to profit or loss

Discount rate: determined by reference to market yields, at the reporting date, on high-quality corporate bonds (or, where no deep market exists for such bonds in the relevant currency, on government bonds in that currency), matching the currency and estimated term of the obligation.

5. Worked Example: The Projected Unit Credit Method

Setup: a lump-sum benefit equal to 1% of final salary per year of service is payable on termination. Year-1 salary = 10,000, growing 7% compound per year. Discount rate = 10%. The employee is expected to leave at the end of Year 5 (no probability adjustment applied, for simplicity).

The build up: salary rises from 10,000 to 10,700, 11,449, 12,250 and 13,108 across Years 1 to 5. Each year earns 1% of the projected final salary, or 131.08. Each period of service therefore adds one unit of benefit measured using the same projected final salary.

Year	1	2	3	4	5
Accrued benefit	131.08	262.16	393.24	524.32	655.40
Years remaining	4	3	2	1	0
Opening obligation	0	89.53	196.96	324.99	476.65
Current service cost	89.53	98.48	108.33	119.16	131.08
Interest cost	0	8.95	19.70	32.50	47.67
Closing obligation	89.53	196.96	324.99	476.65	655.40

Read the calculation in this order:
• Project final salary, then multiply by 1% to obtain the nominal benefit unit of 131.08 for each year of service.
• Multiply this unit by the number of years already worked to obtain the accrued benefit at each reporting date.
• Discount the accrued benefit at 10% over the years remaining until the assumed departure at the end of Year 5. This gives the closing obligation.
• Current service cost is the present value of the new unit earned during the current year. Calculate 131.08 divided by 1.10 raised to the number of remaining years.
• Interest cost is 10% of the opening obligation. Opening obligation plus current service cost plus interest cost reconciles to closing obligation, subject to rounding.

With the same nominal unit and discount rate, current service cost rises as retirement approaches because fewer years of discounting remain. The slide lists 89.53, 98.48, 108.33, 119.16 and 131.08 in Years 1 to 5. Do not reverse this sequence. These figures use the slide's unchanged actuarial assumptions and omit the probability of an earlier or later departure.

Year 2: two units have accrued, giving 262.16. Three years remain, so the closing obligation is 262.16 / 1.10^3 = 196.96. Current service cost is 131.08 / 1.10^3 = 98.48. Interest cost is 89.53 × 10% = 8.95. The roll forward is 89.53 + 98.48 + 8.95 = 196.96.

6. Plan Assets

Plan assets: assets held by a long-term employee benefit fund, or qualifying insurance policies, used to fund the defined benefit obligation.

Must know (plan asset mechanics):
• Measured at **fair value** at each period-end
• Generate **interest income**, computed by applying the **same discount rate** used for the obligation to the plan assets' opening fair value (this interest income reduces the net interest cost in P&L)
• Increase from employer **cash contributions** made during the period
• Any gap between the **actual** return earned and the interest income already credited is a **remeasurement**, recognised in OCI (never recycled), exactly mirroring the obligation-side remeasurement treatment

7. The Five Components of Defined Benefit Accounting

Must know (summary table, component → statement → meaning):
• **1. Service cost** (current + past service/plan amendments) → **P&L** → cost of current/past employee service
• **2. Net interest on the net defined benefit liability/asset** (discount rate × opening net position) → **P&L** → effect of discounting the obligation, net of return on plan assets
• **3. Remeasurements** (actuarial gains/losses, experience gains/losses, actual-vs-expected plan-asset return) → **OCI, never recycled** → long-term changes outside management's control
• **4. Contributions by employer** → reduces the liability (increases plan assets) on the balance sheet → cash outflow into the fund, not itself an expense
• **5. Benefits paid to employees** → reduces both the liability and plan assets → settles part of the obligation, not an expense

Past service cost: arises when the entity must redetermine the liability due to (1) a plan amendment (e.g. a higher benefit percentage) or (2) a plan curtailment/settlement (e.g. following a plant closure or collective redundancy). Past service cost is recognised in P&L at the earlier of: (a) the date the plan change/reduction occurs, or (b) when the entity recognises related restructuring costs (IAS37, Session 14) or termination benefits.

8. Comprehensive Worked Examples

Two-year comprehensive example: opening 01/01/25: obligation 500,000, plan assets 450,000, discount rate 10%.

2025: service cost 150,000. Past service cost 0. Actual return on plan assets 50,000. Remeasurement of the obligation (actuarial changes) 75,000. Benefits paid 80,000. Employer cash contributions 100,000.
Plan assets: opening 450,000 + interest income (10%×450,000=45,000) + remeasurement (actual return 50,000 − interest income 45,000 = 5,000) − benefits paid 80,000 + contributions 100,000 = **520,000** closing.
Obligation: opening 500,000 + interest cost (10%×500,000=50,000) + service cost 150,000 + past service cost 0 + remeasurement 75,000 − benefits paid 80,000 = **695,000** closing.
Net obligation (balance sheet) = 695,000 − 520,000 = **175,000**.
P&L (before OCI): −service cost 150,000 − net interest (interest cost 50,000 − interest income 45,000 = 5,000) = **−155,000**.
OCI: −remeasurement of obligation 75,000 + remeasurement of plan assets 5,000 = **−70,000**. Total comprehensive income effect = −155,000 − 70,000 = **−225,000**.

2026 (opening obligation 695,000, opening plan assets 520,000): service cost 155,000. Past service cost 15,000. Actual return 66,000. Remeasurement of obligation 60,000. Benefits paid 85,000. Contributions 100,000.
Plan assets: 520,000 + interest income (10%×520,000=52,000) + remeasurement (66,000−52,000=14,000) − 85,000 + 100,000 = **601,000**.
Obligation: 695,000 + interest cost (10%×695,000=69,500) + 155,000 + 15,000 + 60,000 − 85,000 = **909,500**.
Net obligation = 909,500 − 601,000 = **308,500**.
P&L: −155,000 − 15,000 − (69,500−52,000=17,500) = **−187,500**. OCI: −60,000 + 14,000 = **−46,000**. Total = −233,500.

Single-year exercise pattern (Exercise 1): opening obligation 800,000, plan assets 450,000, discount rate 4%. Restated obligation from turnover-estimate changes = 870,000 (i.e. a remeasurement of 70,000, computed as the balancing figure once interest cost and service cost are applied). Service cost 70,000. Actual return 42,000. Past service cost 8,000. Contributions 150,000. Benefits paid 40,000.
Plan assets: 450,000 + interest income (4%×450,000=18,000) + remeasurement (42,000−18,000=24,000) − 40,000 + 150,000 = **602,000**.
Obligation: 800,000 + interest cost (4%×800,000=32,000) + 70,000 + 8,000 + remeasurement 70,000 − 40,000 = **940,000**.
Net liability = 940,000 − 602,000 = **338,000**. P&L = −70,000−8,000−(32,000−18,000=14,000) = **−92,000**. OCI = −70,000+24,000 = **−46,000**.

9. Common Mistakes and Exam Traps

- Recognising remeasurement gains/losses in profit or loss. They always go to OCI and are permanently excluded from P&L (never recycled), the same “no recycling” rule seen for equity-FVOCI investments (Session 8).
- Applying different discount rates to the obligation and to plan assets’ interest income. IAS19 requires the **same** rate for both, which is precisely what makes “net interest” a clean, single number.
- Treating employer contributions or benefits paid as P&L expenses. Both are pure balance-sheet movements (contributions increase plan assets, benefits paid reduce both the obligation and plan assets), never income-statement items.
- Confusing “other long-term benefits” (Category 3) with defined benefit post-employment plans (Category 2). Both use similar actuarial mechanics, but other long-term benefits route *all* remeasurement changes through P&L, with no OCI split at all.
- Forgetting that past service cost is recognised immediately (at the earlier of the amendment date or related restructuring recognition), not spread over any future service period. Unlike ordinary service cost, which accrues as service is rendered.
- Mixing up “remeasurement of plan assets” sign convention. It is always (actual return − interest income already credited), so it can be positive (assets outperformed) or negative (underperformed), and either way it is an OCI item, not smoothed into future periods.

10. Must Know Absolutely

Must know:

• Four benefit categories: short-term (undiscounted, as earned), post-employment (this session), other long-term (similar mechanics but all remeasurement to P&L), termination (liability when offer can’t be withdrawn)

• Defined contribution: risk on employee, simple expense-as-due accounting. Defined benefit: risk on employer, full actuarial/discounting model

• Defined benefit liability = projected unit credit method obligation − plan assets (fair value). Deficit = net liability, surplus = net asset

• P&L: service cost + net interest (discount rate × opening net position). OCI (never recycled): all remeasurements (obligation actuarial changes + plan-asset return vs. interest income)

• Same discount rate used throughout: to discount the obligation, and to compute interest income on plan assets

• Past service cost: recognised immediately at the earlier of plan-change date or related restructuring recognition